Ambroise Odonnat

PhD student in ML — defense in March 2027

Paris, France
+33 6 04 53 75 61
ambroiseodonnattechnologie@gmail.com
ambroiseodt.github.io
ambroiseodt
Google Scholar

Summary

I study how transformers learn and generalize, notably under distribution shift, from task-specific models to large language models and vision foundation models. My work combines theory and large-scale experiments, covering supervised training, pretraining, in-context learning, frozen representations, finetuning, and tool use. I also contributed to research on test-time scaling and recursive self-improvement in LLMs. Next, I want to help build and improve frontier models, notably their reasoning capabilities.

Expertise — Science of transformers (representations, training dynamics, theory); distribution shift; unsupervised evaluation.

Publications — ICML (3), NeurIPS (2), ICLR (1), AISTATS (1), ICASSP (1), TMLR (2); 5 as first author.

Impact — Oral at ICML 2024 (top 1.5% of submissions); SAMformer deployed in production.

Research Experience

2024–present **Huawei Noah's Ark Lab**, *Research Scientist Assistant, CIFRE PhD program*, Paris.
Advisors: Prof. Romain Tavenard, Prof. Laetitia Chapel, Dr. Ievgen Redko.
- Derived generalization bounds for LLMs, developed unsupervised accuracy estimators, analyzed transformer representations for finetuning and probing, and studied tool use in LLMs; experiments on models up to 12B parameters.
- **First author of 8 papers**: 3 main conferences (NeurIPS, ICML, ICASSP); 2 workshops (ICLR, ICML); 3 preprints.
- Contributed to recursive self-improvement of agents built on frontier models (IMO proof-grading evaluation).
- **Supervision**: co-supervised a research intern on test-time scaling in LLMs (paper accepted at ICML).

2023–2024 **Huawei Noah's Ark Lab**, *Research Scientist*, Paris.
Advisor: Dr. Ievgen Redko.
- Co-developed SAMformer, a transformer for forecasting (ICML Oral) and an unsupervised accuracy estimator (TMLR).
- **Internal impact**: SAMformer deployed by business units to forecast battery charge cycles.

2023 **Huawei Noah's Ark Lab**, *Research Intern (6 months)*, Paris.
Advisors: Dr. Ievgen Redko, Dr. Vasilii Feofanov.
- Developed a method for robust self-training under sample selection bias (AISTATS).

2022 **Polytechnique Montréal**, *Research Intern (6 months)*, Montréal.
Advisor: Prof. Julien Cohen-Adad.
- Developed a DL pipeline to detect epileptic spikes in EEG (QBIN conference 2022, Best Flash Talk Award).

Education

2024–present **PhD in Computer Science**, *Univ. Rennes 2, Inria, IRISA & Huawei Noah's Ark Lab.*
Understanding Generalization in Transformers: From Specialized to Foundation Models.
Advisors: Prof. Romain Tavenard, Prof. Laetitia Chapel, Dr. Ievgen Redko.

2022–2023 **MSc Mathematics, Vision, Learning (MVA)**, *ENS Paris-Saclay*.
Summa cum laude, GPA 4.0/4.0. Advisor: Prof. Gabriel Peyré.

2019–2023 **Engineering Degree (MSc)**, *École des Ponts ParisTech, Institut Polytechnique de Paris*.
Summa cum laude, GPA 3.9/4.0.

2017–2019 **Preparatory Classes (CPGE)**, *Lycée Henri IV*, MPSI-MP.

Selected Publications (full list)

2026 **Probing Transformer Representations Layer by Layer, Module by Module** — **Preprint**
first author — LLMs, vision foundation models, frozen representations — extension of ICLR Workshop

2026 **The Red Queen Gödel Machine: Co-Evolving Agents and Their Evaluators** — **arXiv**
co-author — LLMs, self-improving agents — **featured in Cambridge CS & State of AI Report 2026**

2026 **Post-Training in Time Series Foundation Models: A Unifying Framework** — **arXiv**
*first author** — time series foundation models, post-training

2026 **Vision Transformer Finetuning Benefits from Non-Smooth Components** — **ICML**
first author — vision foundation models, finetuning

2026 **Optimal Self-Consistency for Efficient Reasoning with LLMs** **ICML**
co-supervisor — LLMs, test-time scaling

2026 **Provable Benefits of In-Tool Learning for Large Language Models** **ICLR Workshop**
*first author** — LLMs, tool use, finetuning

2026 **A Mechanistic Study of Transformers Training Dynamics** **ICML Workshop**
first author — transformers on arithmetic tasks, supervised training

2025 **Easing Optimization Paths: a Circuit Perspective** **ICASSP**
*first author** — transformers on arithmetic tasks, supervised training

2024 **Large Language Models as Markov Chains** **arXiv**
*first author** — LLMs, pretraining, in-context learning — **featured in Forbes**

2024 **MaNo: Exploiting Matrix Norm for Unsupervised Accuracy Estimation** **NeurIPS**
*first author** — distribution shift

2024 **SAMformer: Unlocking the Potential of Transformers in Time Series Forecasting** **ICML Oral**
*first author** — supervised training, generalization — **integrated in PyTorch Forecasting**

2024 **Leveraging Ensemble Diversity for Robust Self-Training under Sample Selection Bias** **AISTATS**
first author — supervised training, distribution shift

*Equal contribution.

Awards & Grants

2023 **Merit Scholarship**, Chair Saint-Gobain & École des Ponts ParisTech ($5,800)

2022 **Research grant**, proposal author, Tanenbaum Open Science Institute & McGill University ($10,800)

2022 **Best Flash Talk Award**, QBIN Conference ($1,000)

Invited Talks (full list)

Sep. 2026 **Dauphine** — A mechanistic study of transformers training dynamics

Mar. 2026 **Mila** — Vision transformer finetuning benefits from non-smooth components

Mar. 2026 **Imperial College London** — Provable benefits of tool use for LLMs

Feb. 2026 **Kyutai** — Provable benefits of tool use for LLMs

Oct. 2025 **Cohere** — Provable benefits of tool use for LLMs

Jun. 2025 **Cohere Labs** — Large language models as Markov chains

Feb. 2025 **EPFL** — Large language models as Markov chains

Sep. 2024 **ENS Ulm** — SAMformer: transformers for time series forecasting

Jul. 2024 **Criteo AI Lab** — SAMformer: transformers for time series forecasting

Community Service & Teaching

2026 **Conference Organizer**, *ELLIS UnConference*, European ML gathering, ∼2000 participants.

2025 **Conference Organizer**, *NeurIPS@Paris*, Sorbonne University, ∼350 participants.

2025 **Workshop Organizer**, *NeurIPS 2025*, BERT^2: Recent Advances in Time Series Foundation Models.

Reviewing NeurIPS (2025), ICML (2024–2026), ICLR (2025, 2026), AISTATS (2026), TMLR

Teaching ML teaching assistant (M2), École des Ponts (2024); maths examiner for HEC-ESSEC entrance exam (2024).

Skills

ML stack Python, PyTorch, Hugging Face Transformers & PEFT, cuML, NVIDIA Brev

Open source Reproducible code for my papers (GitHub); contributor to skada, skada-bench, and benchopt.

Languages French (native), English (fluent), German (beginner)

Other Model United Nations in Rome (2015); fencing (10 years), university football & volleyball.

References

Charles Arnal, FAIR at Meta
charles.a.arnal@gmail.com

Vivien Cabannes, FAIR at Meta
vivien.cabannes@gmail.com

Laetitia Chapel, L'Institut Agro, IRISA
laetitia.chapel@irisa.fr

Amaury Hayat, École des Ponts ParisTech
amaury.hayat@gmail.com

Ievgen Redko, Schneider Electric
ievgen.redko@gmail.com

Romain Tavenard, Univ. Rennes 2, Inria
romain.tavenard@univ-rennes2.fr